Summary in Graph

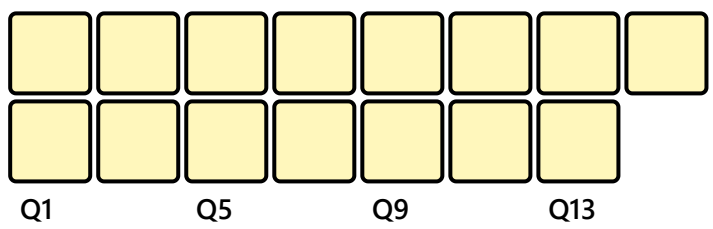
Exam Summary (GO Classes CS Test Series 2026 | Algorithms | Topic Wise Test 4)

Qs. Attempted:	0 0 + 0	Correct Marks:	0 0 + 0
Correct Attempts:	0 0 + 0	Penalty Marks:	0 0 + 0
Incorrect Attempts:	0 0 + 0	Resultant Marks:	0 0 + 0

Total Questions:	15 5 + 10
Total Marks:	25 5 + 20
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (66) out of 660 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Numerical Type Award: 1 Penalty: 0 Algorithms

Consider the following recurrence relation which is applicable on two arrays x and y. x_i and y_i are the ith elements of x and y array respectively.

$$c(i,j) = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ c(i-1, j-1) + 1 & \text{if } i, j > 0 \text{ and } x_i = y_j \\ \min(c(i-1, j), c(i, j-1)) & \text{if } i, j > 0 \text{ and } x_i \neq y_j \end{cases}$$

Suppose each array is of n size and time complexity to compute $c(n,n)$ using dynamic programming is $O\left(n^a(\log n)^b\right)$ then what will be value of $a+b$?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 03sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

You are working on a dynamic programming problem defined by the recurrence:

$$A(i,j) = F\big(A(\lfloor i/2 \rfloor,\, j),\ A(i,\, \lfloor j/2 \rfloor)\big),$$

where F can be evaluated in constant time.

$$A(i,j) = 0 \quad \text{whenever } i = 0 \text{ or } j = 0.$$

To compute $A(m,n)$, you may use either of the following approaches:

- Top-down dynamic programming (memoized recursion starting from $A(m,n)$).
- Bottom-up dynamic programming (filling a table of size $(m+1)\times(n+1)$, beginning from $A(0,0)$).

Which method is asymptotically more efficient?

A. Top-down will take asymptotically less time than bottom-up
B. Bottom-up will take asymptotically less time than top-down
C. Both will take the same time
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

In bottom-up dynamic programming, we need an order to fill in the solution cells in a table, such that all needed subproblems are solved before solving a subproblem. For the following relation, give a valid traversal order.

$$\begin{aligned}&\mathrm{A}(i,j) = \mathrm{F}(\mathrm{A}(i,j-1), \mathrm{A}(i-1,j-1), \mathrm{A}(i-1,j+1))\\&\mathrm{A}(0,j) = 0\\&\mathrm{A}(i,0) = 0\end{aligned}$$

Where $\mathrm{F}(.)$ is a function.

A. Fill values of $\mathrm{A}(i,j)$ column-wise, i.e., fill the first column then the second column, and so on.
B. Fill the last column of $\mathrm{A}(i,j)$, then the second last column, and so on.
C. Fill values of $\mathrm{A}(i,j)$ row-wise i.e. fill the first row then the second row, and so on.
D. No order is possible as there is cyclic dependency in subproblems.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let $X = x_1x_2\cdots x_m$ and $Y = y_1y_2\cdots y_n$ be two strings over the alphabet $\Sigma = \{\mathrm{A},\mathrm{C},\mathrm{G},\mathrm{T}\}$
A longest common subsequence (LCS) of X and Y can be found by dynamic programming. 'To compute $c[i,j]$, where $1 \le i \le m$ and $1 \le j \le n$, how many table entries are examined in the worst case?

Where $c[i, j]$ is the length of LCS of $x_1x_2 \cdots x_i$ and $y_1y_2 \cdots y_j$.

A. 2
B. 3
C. n
D. $n + m$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following is/are Longest Common Subsequence for given two sequecnces $\langle 1, 0, 0, 1, 0, 1, 0, 1 \rangle$ and $\langle 0, 1, 0, 1, 1, 0, 1, 1, 0 \rangle$.

A. $\{1, 0, 0, 1, 1, 0\}$.
B. $\{1, 0, 1, 0, 1, 1\}$
C. $\{1, 0, 1, 1, 0, 1\}$
D. $\{0, 0, 1, 1, 1, 0\}$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider two teams, A and B, playing a series of games until one of the teams wins n games. Assume that the probability of A winning a game is the same for each game and equal to p and the probability of A losing a game is $q = 1 - p$. (Hence, there are no ties.)

Let A needs i more games to win the series and B needs j more games to win the series then $\mathrm{P}(i, j)$ represents the probability of A finally winning the series.

Choose the correct recurrence relation for $\mathrm{P}(i, j)$.
(Hint: If A wins the game then A will need $i - 1$ more wins to win series)

A. $\mathrm{P}(i, j) = p\mathrm{P}(i - 1, j) + q\mathrm{P}(i, j - 1)$
B. $\mathrm{P}(i, j) = p\mathrm{P}(i - 1, j - 1)$
C. $\mathrm{P}(i, j) = p\mathrm{P}(i, j - 1) + q\mathrm{P}(i - 1, j)$
D. $\mathrm{P}(i, j) = q\mathrm{P}(i - 1, j - 1)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following $0 - 1$ knapsack problem with the item's weight and value given in the table.

item	weight	value
1	3	₹25
2	2	₹20
3	1	₹15
4	4	₹40
5	5	₹50

Capacity $\mathrm{W} = 6$.

If the capacity of the knapsack (W), is 6 then how many optimal solutions (number of set of items) are possible?

A. 1
B. 2
C. 3
D. 4

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

For a given sequence of integers $a_1, a_2, \ldots, a_n$, a decreasing subsequence is one for which every integer is strictly smaller than the previous one.

The longest decreasing subsequence is the longest among all decreasing subsequences of a given sequence.
For example, decreasing subsequences of $5, 2, 4, 8, 3, 40$ are -

- $5, 2$
- $5, 4$
- $4, 3$
- $5, 4, 3$ etc

And longest among all possible decreasing subsequences is $5, 4, 3$.
For a given sequence of integers $a1, a2, \ldots, a_n$, let $\mathrm{L}(j)$ be length of the longest decreasing subsequence that starts with a_j.

$$\mathrm{L}(n) = 1$$

$$\mathrm{L}(j) = \begin{cases} 1 & \text{if } a_j = \min\left(a_j, a_{j+1}, a_{j+2}, \ldots, a_n\right), \\ ? & \text{otherwise.} \end{cases}$$

Complete the above recurrence relation with the appropriate value of $?$.

A. $1 + \min\{\mathrm{L}(k) : j < k \leq n \text{ and } a_j > a_k\}$
B. $1 + \max\{\mathrm{L}(k) : j < k \leq n \text{ and } a_j > a_k\}$
C. $1 + \max\{\mathrm{L}(k) : j < k \leq n \text{ and } a_j < a_k\}$
D. $1 + \min\{\mathrm{L}(k) : j < k \leq n \text{ and } a_j < a_k\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider a DAG $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ which has topological ordering of vertices $v_1, v_2, \ldots, v_n$. We want to count number of paths from vertex v_i to vertex v_n. Let $\mathrm{Paths}(i)$ represents total number of paths from i to n.

Complete the following recurrence for $\mathrm{Paths}(i)$.

$$\mathrm{Paths}(i) = \begin{cases} 1 & \text{if } i = n \\ ? & \text{otherwise.} \end{cases}$$

In options,

$\sum_{j:(i,j)\in\mathrm{E}} \mathrm{Paths}(j)$ is a quantity which sums $\mathrm{Paths}(j)$ for all outgoing edges from i i.e., for edges $i \to j$.

A. $\sum_{j:(i,j)\in\mathrm{E}} (1 + \mathrm{Paths}(j))$

B. $\sum_{j:(i,j)\in \mathrm{E}} \mathrm{Paths}(j)$
C. $1+\sum_{j:(i,j)\in \mathrm{E}} \mathrm{Paths}(j)$
D. $1+\mathrm{Paths}(j)$ for some edge $i \rightarrow j$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Find total number of scalar multiplications of a matrix-chain product of 6 matrices whose sequence of dimensions is $5, 10, 3, 12, 5, 50$ and 6. That is 5×10 is dimension of first matrix, 10×3 is dimension of second matrix and so on.

Your Answer: | Correct Answer: 2010 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

The number of longest common subsequences for "$bacb$" and "$abcabc$" are -

A. 2
B. 3
C. 4
D. 5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

For the given recurrences calculate the tightest bound on time and space that would be required by a dynamic programming algorithm to compute $\mathrm{OPT}(n)$.

$$\mathrm{OPT}(i) = \min_{1\leq j<i} \{\mathrm{OPT}(j)/j + w(j)\}.$$

where $\mathrm{OPT}(1) = 1$

A. Time $= O\left(n^2\right)$, space $= O(n)$.
B. Time $= O\left(n^3\right)$, Space $= O\left(n^2\right)$.
C. Time $= O\left(n^2\right)$, Space $= O\left(n^2\right)$.
D. Time $= O(n)$, Space $= O(1)$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose you have a row of coins with values that are positive integers $c_1, \ldots, c_n$. These values might not be distinct. Your task is to pick up coins that have as much total value as possible, subject to the constraint that you don't ever pick up two coins that lie beside each other. For example - Given c_1 to c_6 as follows: $(5, 1, 2, 10, 6, 2)$, the maximum value is 17 and using coins $\{c_1 = 5, c_1 = 10, c_6 = 2\}$.

Let $f(n)$ be maximum total value with n coins with $f(0) = 0$ and $f(1) = c_1$ then what will be correct recurrence for $f(n)$?

A. $f(n) = \max(c_n + f(n-2), f(n-1))$
B. $f(n) = \max(c_{n-1} + f(n-2), f(\mathrm{n}-1))$
C. $f(n) = \max(c_{n-1} + f(n-1), f(n-2))$
D. $f(n) = \max(c_n + f(n-1), f(n-2))$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

For $x \geqq 0, y \geqq 0$, define $A(x, y)$ by

$$\begin{aligned} &A(0, y) = y + 1 \\ &A(x+1, 0) = A(x, 1), \text{ and} \\ &A(x+1, y+1) = A(x, A(x+1, y)) \end{aligned}$$

Then, for all non-negative integers y, $A(1, y)$ is _____

A. 2
B. $y + 1$
C. $y + 2$
D. $2y + 3$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

An alignment between two strings w_1 and w_2 (over the alphabet $\{a, b, c, \ldots, z\}$) is obtained by inserting hyphens in the two strings such that the modified strings align. A mismatch in an alignment is a position where one of the modified strings has a hyphen, and the other does not.

$$\text{Oc-urr-ance}$$

$$\text{Occurre-nce}$$

There are three mismatches in this example.

Let $s[1 \ldots m]$ and $t[1 \ldots n]$ be the two strings to be matched. Let $\mathrm{M}(i, j)$ be the number of mismatches in the best alignment of $s[1 \ldots i]$ and $t[1 \ldots j]$.

Which of the following is a correct recursive formulation of $\mathrm{M}(i, j)$ for $i, j > 0$?

Note: By convention $s[1 \ldots 0]$ and $t[1 \ldots 0]$ are taken to be the empty string, so $\mathrm{M}(i, 0) = i$ for $i \in \{1, 2, \ldots, m\}$ and $\mathrm{M}(0, j) = j$ for $j \in \{1, 2, \ldots, n\}$

A. $\mathrm{M}(i, j) = \begin{cases} 2 + \mathrm{M}(i-1, j-1), \text{if } s[i] \neq t[i] \\ \min(\mathrm{M}(i-1, j), \mathrm{M}(i, j-1)), \text{ otherwise} \end{cases}$
B. $\mathrm{M}(i, j) = \begin{cases} \mathrm{M}(i-1, j-1), \text{if } s[i] = t[j] \\ \min(\mathrm{M}(i-1, j), \mathrm{M}(i, j-1)), \text{ otherwise} \end{cases}$
C. $\mathrm{M}(i, j) = \begin{cases} 1 + \mathrm{M}(i-1, j-1), \text{if } s[i] = t[j] \\ 1 + \min(\mathrm{M}(i-1, j), \mathrm{M}(i, j-1)), \text{ otherwise} \end{cases}$
D. $\mathrm{M}(i, j) = \begin{cases} \mathrm{M}(i-1, j-1), \text{if } s[i] = t[j] \\ 1 + \min(\mathrm{M}(i-1, j), \mathrm{M}(i, j-1)), \text{ otherwise} \end{cases}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff